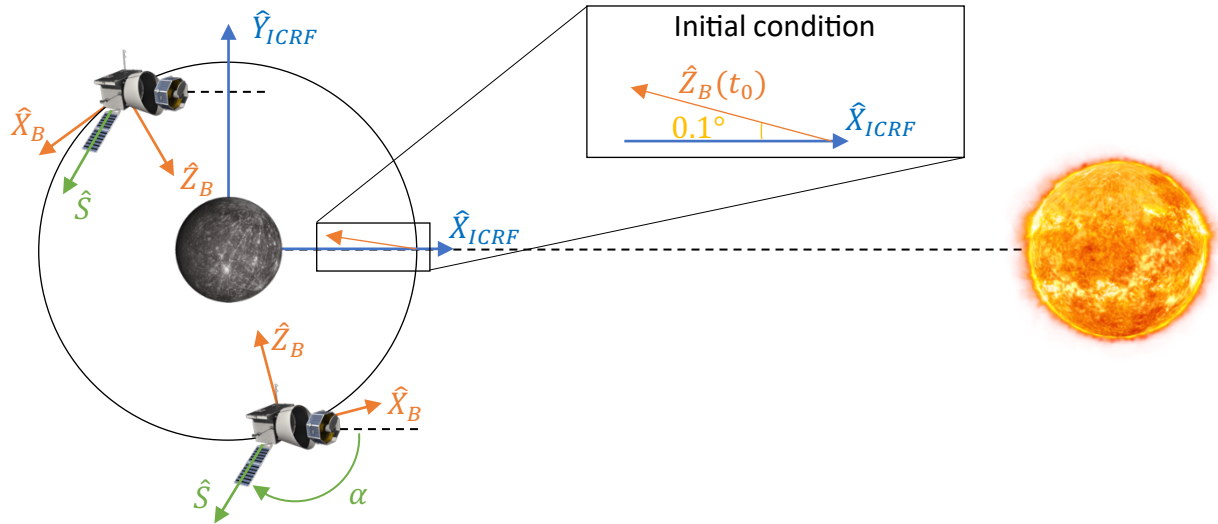


Space Missions and Systems 2026: Homework #2

The **BepiColombo** mission will arrive at Mercury in November 2026. The spacecraft will be inserted into a circular, polar orbit, with orbital radius of **3430 km**. The initial geometry is shown in the figure below.



Due to power constraints, the orientation of the solar panels is fixed with respect to the Sun direction. The distance between Mercury and the Sun is assumed to be $D = 0.41 \text{ AU}$. Mercury may be considered fixed in space and non-rotating, with the Sun direction aligned with the $\hat{X}_{ICRF}$ axis.

The spacecraft body frame is shown in the figure. In the nominal nadir-pointing attitude, the $\hat{Z}_B$ axis points toward Mercury's center, while the $\hat{X}_B$ axis is aligned with the along-track direction. At $t = t_0$, the spacecraft is not exactly nadir-pointing: the $\hat{Z}_B$ axis has an initial offset with respect to the nadir direction, as shown in the figure. The spacecraft frame is initially not rotating in the ICRF.

The spacecraft is subject to a disturbance torque caused by solar radiation pressure. The gravity gradient can be neglected. The centre of pressure lies along the $\hat{S}$ axis (along the solar panel direction), at 2 meters from the origin of the spacecraft body frame. The angle between $\hat{S}$ and $\hat{X}_{ICRF}$ is kept constant and equal to $\alpha = -120^\circ$. When the spacecraft is in eclipse, the solar radiation pressure disturbance torque does not act on the spacecraft.

Consider an active attitude control system using four momentum wheels mounted in a tetrahedral configuration. The tasks are:

1. Using the first three momentum wheels, design an attitude control system able to bring the spacecraft to the nadir-pointing condition with a tolerance of $\pm 4 \text{ arcsec}$, within 5 minutes.
2. Determine for how long the momentum wheels can maintain the desired nadir-pointing attitude.
3. Using the same initial conditions and same geometry, repeat points 1 and 2 assuming now that the orientation of the solar panel orientation is fixed in the spacecraft body frame, with an offset of -100° with respect to the $\hat{X}_B$ axis.

Read carefully the following information:

- Mercury radius: 2440 km
- Mercury GM : 22032 km³/s²
- Solar flux at 1 AU (ϕ): 1371 W/m²
- Speed of Light (c) = 299792.458 km/s
- Solar panel area (A): 8 m²
- Solar panel reflectivity specular coefficient (C_s): 0.3
- The SRP force is equal to:

$$\vec{F}_{SRP} = -\frac{\phi}{c} \left(\frac{1 \text{ AU}}{D_{MPO}} \right)^2 A \hat{n} \cdot \hat{X}_{ICRF} \left((1 - C_s) \hat{X}_{ICRF} + 2C_s (\hat{n} \cdot \hat{X}_{ICRF}) \hat{n} \right)$$

Where $\hat{n}$ is the normal of the solar panel surface looking at the Sun. **Pay attention to which side of the solar panel faces the Sun.**

- Spacecraft moment of inertia: 6500 kg m²
- The mounting matrix is defined as follow:

$$\begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} T_{c,1} \\ T_{c,2} \\ T_{c,3} \\ T_{c,4} \end{bmatrix}$$

- Wheels moment of inertia: 0.054 kg m²
- Wheels maximum angular speed: ± 4000 rpm (revolutions per minute)
- Wheels maximum torque: ± 0.511 Nm
- Wheels initial angular velocity: [1751, -121, -1000, 0] rpm
- The electric equations of the reaction wheels are:

$$i_W = \frac{T_C}{K_M}, \quad P_W = V_M i_W = (K_W \omega_W + R i_W) i_W$$

- $K_M = 0.118$ Nm/A
- $K_W = 0.008$ V/rpm
- $R = 1.2 \Omega$
- $P_{max} = 29$ W

Upload on Google Classroom a **zip folder** containing a working computer code and instructions for its use. The code must run with a one-line command. It must print tables and produce the relevant plots (MATLAB is recommended). You must provide a **comprehensive and concise note in English** (a separate pdf file **not included in the zip folder**) containing **all** mathematical procedures, results, comments, tables and figures. **Deadline: Sunday 17 May 23:59:59. No exceptions.**

Only questions pertaining to the text will be answered. Please ask questions publicly on Classroom (in English).